

# Miguel L. Rodrigues

Computer Vision & MLOps Engineer

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Computer vision engineer building **production deep-learning systems** for industrial inspection, from dataset curation and model development to optimized cloud inference and monitoring. Hands-on with **Python, PyTorch, OpenCV, AWS, Docker, ONNX, and INT8**; M.Sc. researcher with peer-reviewed publications in robust control.

## ■ EXPERIENCE

**AI / Computer Vision Engineer** Oct 2024 – Present  
Digimet Solutions — Joinville, SC (remote)

- Develop and maintain a production **weld-bead inspection** pipeline in Python/PyTorch: segmentation and keypoint models feeding geometric measurement.
- Own the model lifecycle end to end: dataset curation and annotation infrastructure (Label Studio · EC2 · PostgreSQL · S3), training, validation, and mask post-processing with OpenCV.
- Optimize and deploy serverless inference on **AWS Lambda**; use ONNX and INT8 quantization for latency and cost, CloudWatch for monitoring, and domain adaptation for field imagery.

**Undergraduate Researcher** (scholarship & volunteer) Feb 2021 – Mar 2024  
CEFET-MG — Departments of Mechatronics & Computer Engineering

- Proposed, implemented, and evaluated algorithms from scientific literature: controller synthesis, Lyapunov analysis, Takagi–Sugeno neuro-fuzzy modeling, and AI-based control of manipulators (KUKA youBot).
- Applied fuzzy inference, ML, and image processing to investment recommendation, virtual drug screening, and prostate-brachytherapy automation.

**Teaching Assistant** Mar – Sep 2024  
CEFET-MG — Signals & Systems; Control Theory

## ■ RESEARCH & PUBLICATIONS

|                                                                                     |                                                                                                                                                           |             |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|  | Event generators that cut transmissions in networked loops while certifying stability — LPV models, saturation, LMIs, evolutionary co-design.             |             |
| EVENT-TRIGGERED<br>& ROBUST CONTROL                                                 | Gerador de Eventos Adaptativo com Rastreamento de Função Hazard para Sistemas LPV Saturantes. <b>Rodrigues</b> , Silva, Leite.                            | CBA 2026    |
|                                                                                     | Co-Design and Evolutionary Optimization of Periodic Event-Triggered Robust Control for LPV Systems. Oliveira, <b>Rodrigues</b> , Guelton, Motchon, Leite. | ROCOND 2025 |
|                                                                                     | Gerador Dinâmico de Eventos com Tratamento de Ruído para Redução de Transmissões. <b>Rodrigues</b> , Silva, Leite.                                        | SBAI 2025   |
|                                                                                     | Gerador de Eventos para Minimização de Transmissões em Sistemas LPV Saturantes. <b>Rodrigues</b> , Silva, Leite.                                          | CBA 2024    |
|  | Modelagens e Simulações Computacionais Aplicadas ao Desenvolvimento Remoto de Sistemas Mecatrônicos. Silva, Alves, <b>Rodrigues</b> , Dâmaso.             | CBA 2022    |
| MECHATRONICS &<br>SIMULATION                                                        |                                                                                                                                                           |             |

## ■ RELEVANT PROJECTS

|                                                                                     |                                                                                                                                                                                               |  |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <b>robotic_tools</b> — serial manipulators from Denavit–Hartenberg parameters: forward/inverse kinematics, Euler–Lagrange dynamics, Jacobians, gradient-based and evolutionary IK.            |  |
| ROBOTICS &<br>MECHANICS                                                             |                                                                                                                                                                                               |  |
|  | <b>prometheus</b> — ESP32 temperature loop with UFMG's GREa group: discrete PID with anti-windup, 2nd-order IIR pre-filter, open-to closed-loop handover, dual-core FreeRTOS, MQTT telemetry. |  |
| EMBEDDED CONTROL                                                                    |                                                                                                                                                                                               |  |

## ■ EDUCATION

**M.Sc., Electrical Engineering** (in progress) 2025 – 2027  
CEFET-MG — Event-triggered control of LPV systems with actuator saturation

**B.Eng., Mechatronics Engineering** 2020 – 2025  
CEFET-MG — Control systems, automation, robotics, embedded systems

## ■ SKILLS & LANGUAGES

|                 |                                                                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| VISION & AI     | Python · PyTorch · OpenCV · image processing · semantic/instance segmentation · keypoint detection · model training, validation & inference |
| CONTROL         | LPV systems · event-triggered & robust control · LMIs · Lyapunov stability · NumPy · SciPy · SymPy                                          |
| MLOPS & CLOUD   | AWS (Lambda, EC2, S3, CloudWatch) · Docker · ONNX · INT8 quantization · CI · Git · Linux                                                    |
| SOFTWARE & EDGE | Python · C · TypeScript · ESP32 · STM32 · ESP-IDF · FreeRTOS · MQTT                                                                         |
| LANGUAGES       | Portuguese (native) · English (full professional) · French (elementary)                                                                     |